

10.6 Energy supply and demand

Globally, energy production and consumption continue to increase. The reasons for this vary, from the continued increase in population, to economic growth, and technological developments.

As population continues to increase, there is a natural increase in the demand for electricity. In addition, the standard of living of the population is improving globally. This results in people having more electrical appliances in their homes, running and heated water, and access to private vehicles. All of these factors increase the energy consumption per capita (per person) globally, further increasing energy demand.

Economic growth also drives an increase in energy demand. As an economy expands, so does industry, which uses energy-using machinery to complete tasks. As outlined above, an improving economy results in an increase in the individual wealth of the population. This means more people own items like cars, further increasing energy demand.

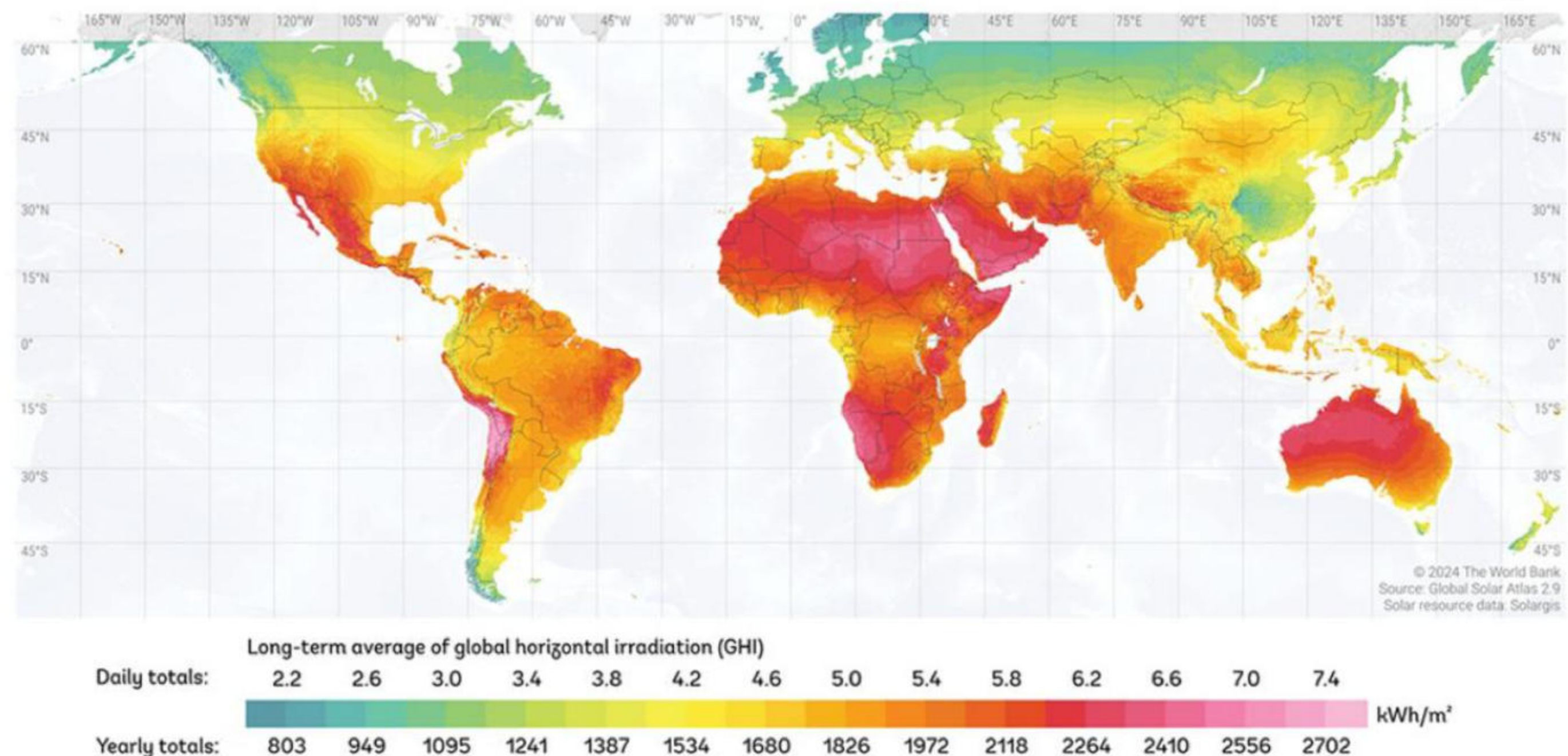
Technological developments have also resulted in the development of a wide range of devices and systems that require energy. This includes personal technology from mobile phones and laptops to smart homes and advanced industry. The increased use of technology by individuals and countries has resulted in an increased demand for energy.

The challenge of supplying energy

The issue of providing sufficient energy is a global challenge. Meeting the ever-increasing demand in a sustainable manner requires a shift in the approach to supplying energy. According to the Energy Information Administration, the world **primary energy demand** is expected to increase by 50% between 2020 and 2050. To meet this demand, the world's electricity-generating capacity has to increase significantly.

Meeting this demand is more challenging for some countries than others. The location of both renewable and non-renewable energy resources has resulted in some regions having a surplus of energy, while other regions suffer from a deficit (shortages).

Some countries have a natural supply of oil and gas, which gives them access to inexpensive energy and a surplus that they can sell to other countries. Other countries have very limited access to fossil fuels and rely on importing energy from other countries. This makes them less energy secure. Renewable energy sources are also regionally dependent. For example, solar power is most effective in regions with high rates of sunlight, wind power requires consistent wind, and geothermal power requires tectonic activity bringing heat close to the surface of the Earth. This means that not all renewable energy sources are suitable for all locations, limiting their use for some countries. Pressure on countries to change to their energy supply from non-renewable to renewable resources further complicates this energy supply challenge for many countries. Governments need to consider how to develop their energy supply to ensure it is secure and meets their needs both now and in the future.



Source **A**: The variation in daily and yearly solar radiation patterns around the world

- a In pairs, discuss which regions are most likely to support the use of solar power as a reliable energy source. Briefly give your reasoning, using information from the source to support your answer.
- b Would the UK be able to depend on solar power to meet any energy deficits? Consider the information in the source and how climate and seasonal changes would influence their ability to rely on solar power.

Energy security

Energy security is the uninterrupted availability of energy sources at affordable prices. Energy security has both short-term and long-term needs. For the short-term, the energy supply needs to meet the current demand for energy and react if there is a sudden change in supply or demand. For example, if a power generator fails, or if there are increases in demand for cooling or heating due to a change in temperature. Long-term security refers to long-term planning to develop energy resources to meet future demand. As populations and economies expand, so does the demand for energy. This means that countries need to consider their rate of growth and the potential demands for energy in the future, ensuring an ongoing development of energy-supply strategies. If a long-term plan is not well designed, countries will find that they are not able to continue to develop their economy in the future.

Without energy security, the economy will struggle to expand, as industry is unable to function fully. This will result in businesses, industry, and commercial farming having to either reduce their output or close, and people losing their jobs and access to a secure supply of food. This increases poverty and decreases the standard of living and quality of life of the population. In some instances, the lack of power supply could pose a direct threat to life if people can't heat their homes in cold climates or cool homes in heatwaves. In addition, the lack of electricity results in hospitals not being able to function properly, water supplies failing, and population health declining.

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Source B: Cartoon depicting a problem arising out of energy insecurity

TASK 2: Study Source B

In pairs, discuss what this cartoon is depicting. Then, in 100 to 200 words, write down the message of the cartoon and explain the possible reasons for such a situation developing.

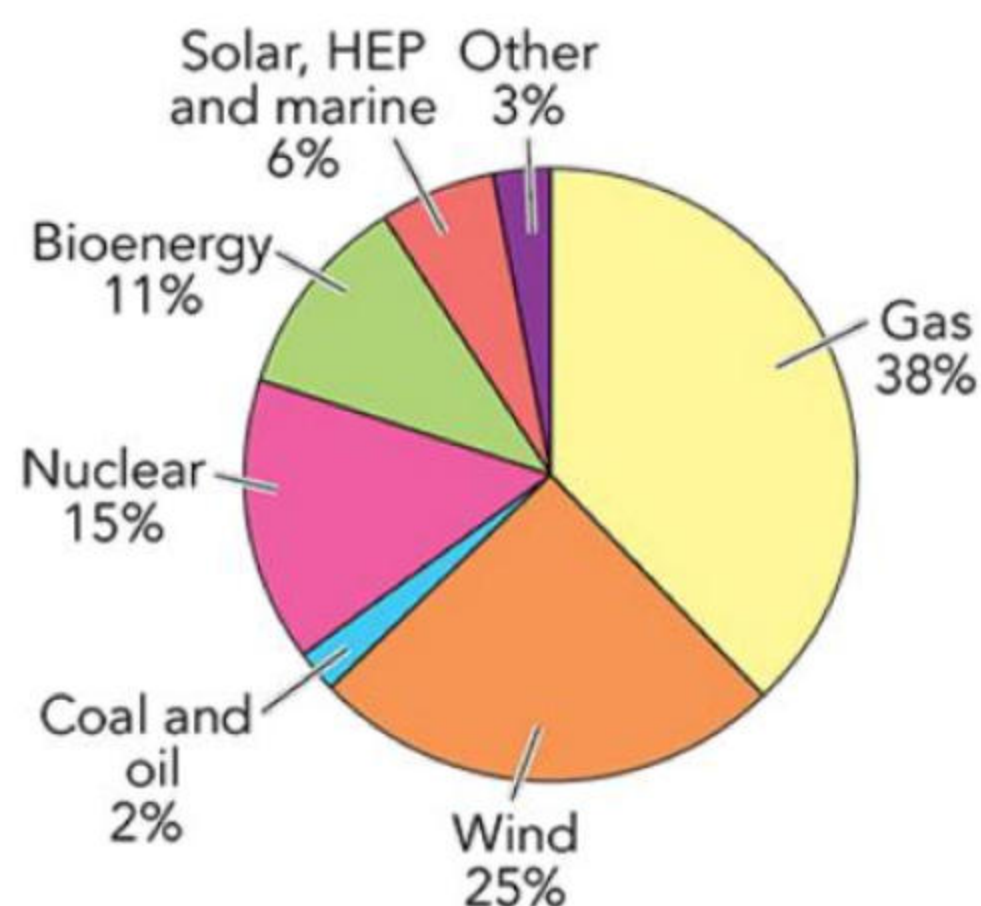
The energy mix

The **energy mix** for countries at different economic levels varies. This is due to a number of factors:

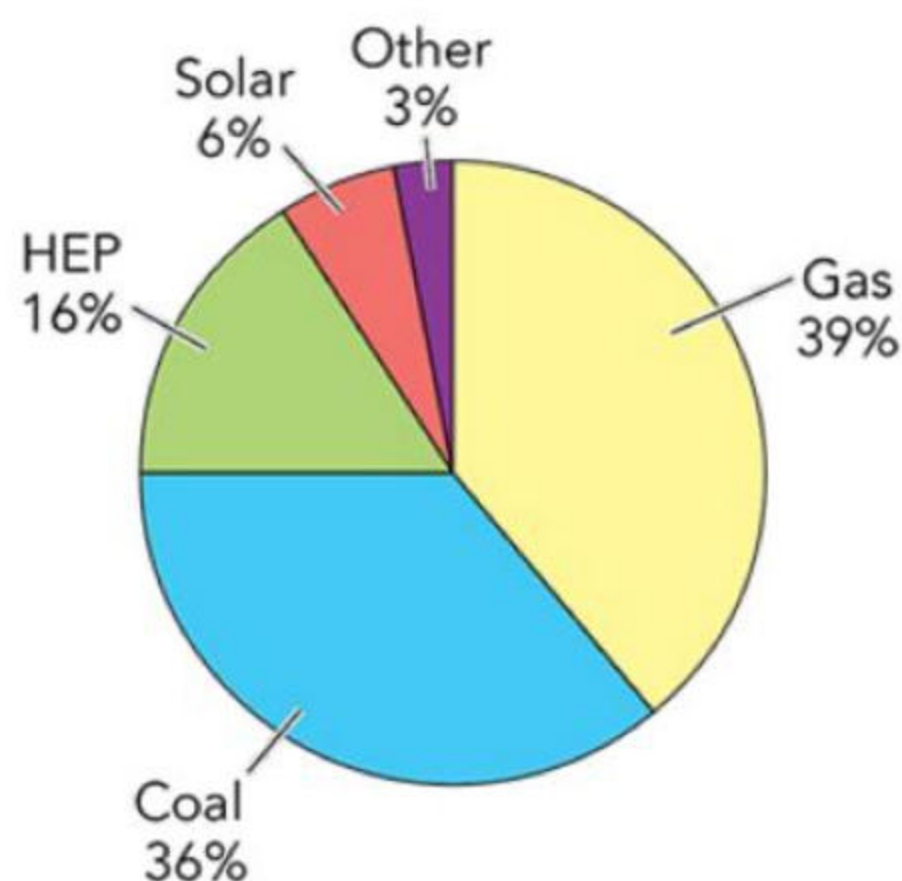
- HICs can invest in a more diverse mix of energy sources than LICs. Developing renewable sources of energy, for example, requires a large investment.
- Energy sources require infrastructure such as transport routes and power for effective installation. Many LICs do not have this infrastructure in place.
- The installation and running of energy sources require specific skills. Often, LICs do not have these highly trained professionals available.
- Political instability and political willingness can also play a role in limiting the development of renewable energy sources. Unless the government is stable and wishing to make the change, it is unlikely to happen.

The energy mix can also vary within countries; for example, along a coastline, a region may be able to develop energy supplies that use wind and tidal energy, while inland areas may be able to develop wind and hydroelectric energy supplies.

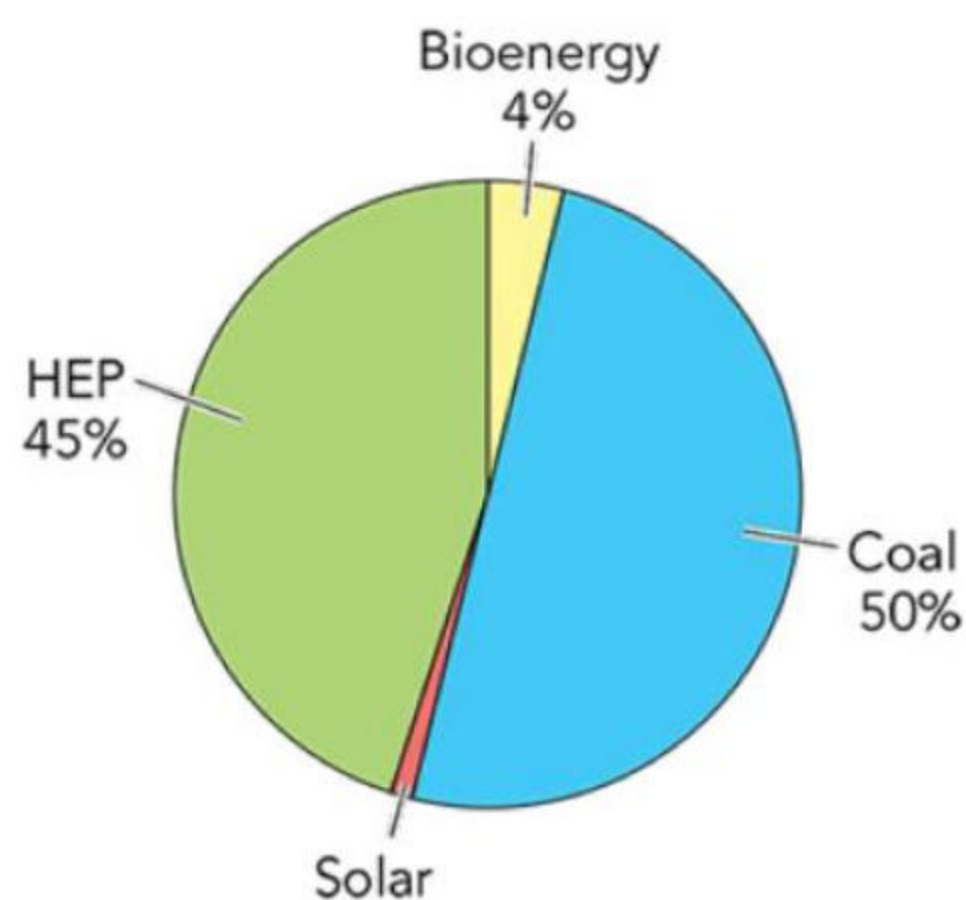
Parts of the country with higher population densities may have their energy supplied by nuclear power, while more remote areas may still rely on fossil fuels as an energy source.



Source **C**: The energy mix for the UK in 2022 (HIC)



Source **D**: The energy mix for Malaysia in 2022 (MIC)



Source **E**: The energy mix for Zimbabwe in 2022 (LIC)

TASK 3: Study Sources **C** **D** **E**

- Which country** has the highest reliance on coal?
- Which country** has the most diverse energy mix?
- Compare the energy mixes** between the UK, Malaysia, and Zimbabwe.
- Suggest why these energy mixes are so different between the three countries.
- In pairs or small groups, carry out research into the mix of energy of your own country, giving reasons for the mix that is used.

TIP

When comparing graphs, take note of the data that is given, and use the data in your answers. Think about how you are using data to support your answers; it should help to clarify what you are trying to say.